Lecture 17

LLM Agents and Systems

COMP532 – Machine Learning and Bioinspired Optimisation

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From Models to Agents

Traditional ML system:

- input $\rightarrow$ model prediction $\rightarrow$ output
- Example: spam classifier
- email $\rightarrow$ model $\rightarrow$ spam / not spam

Agent systems introduce:

- reasoning
- planning
- actions
- Example: autonomous driving

Agent loop:

$$\textit{observe} \rightarrow \textit{decide} \rightarrow \textit{act}$$

What is an LLM Agent?

An LLM agent is a system where a language model:

- interprets instructions
- decides actions
- interacts with tools
- updates context

Example task:

- User: "Find the cheapest flight from London to Paris tomorrow."
- Agent: search flight API $\rightarrow$ compare prices $\rightarrow$ return result

Why LLMs Enable Agents

Large language models provide key capabilities:

- instruction following
- reasoning and planning
- code generation
- natural language interaction

These capabilities allow agents to:

- decide actions
- call tools and APIs
- solve multi-step tasks

Agent Interaction Loop

Agent workflow:

1. Observe context
2. Reason about the task
3. Take an action
4. Receive feedback

This loop repeats until the task is solved.

Planning in LLM Agents

Many tasks require multiple steps.
Example: travel planning

1. find flights
2. check hotel prices
3. compare options
4. generate itinerary

Agents perform:

- task decomposition
- sequential reasoning
- decision making

Reasoning and Acting (ReAct)

The ReAct paradigm combines:

- reasoning steps
- external actions

Example structure:

- Thought
- Action
- Observation

Example:

- Thought: I need the population of France
- Action: Search("population of France")
- Observation: 67 million

Example: Question Answering

User question:

`What is the population of the capital of France?`

Agent reasoning:

- Thought: The capital is Paris
- Action: Search("population of Paris")
- Observation: 2.1 million

Answer:

- The population of Paris is about 2.1 million.

Tool Use

LLM agents interact with external tools:

- search engines (retrieve information)
- databases (query structured data)
- calculators (perform arithmetic)
- code interpreters (execute programs)

Example:

- Question: "What is 173×48?"
- Agent calls a calculator tool
- Returns the computed result

Structured Tool Calls

Agents often generate structured outputs:

- JSON
- function calls
- API requests

The system executes the tool and returns the result to the language model.

Memory in Agents

Agents require memory to maintain context.
Types of memory:

- short-term memory (context window)
- long-term memory (vector databases)

Memory enables agents to:

- remember previous interactions
- store retrieved knowledge
- maintain long tasks

LLM Agent System Architecture

Modern AI agents combine several modules:

- language model (reasoning engine)
- planning module
- tool interface
- memory system

Agents integrate these components to interact with environments and solve tasks.

LLM Agent Architecture

User Query

↓

Language Model
(reasoning and planning)

↓

Tools
Search Database Calculator

↓

Environment Feedback

↓

Memory Update

Example: Coding Agent

Task: write and debug a program.
Agent workflow:

1. generate code
2. run the program
3. observe errors
4. revise the code

Examples include code assistants and automated debugging tools.

Example: Research Assistant

Goal: produce a research report.
Pipeline:

1. plan topics
2. search papers
3. summarize documents
4. generate report

Agents can automate parts of literature exploration.

LLM Agents vs RL Agents

Reinforcement learning agents:

$$\textit{state} \rightarrow \textit{action} \rightarrow \textit{reward}$$

Goal:

- maximize cumulative reward

LLM agents:

$$\textit{context} \rightarrow \textit{reasoning} \rightarrow \textit{action}$$

Goal:

- solve tasks via reasoning and tool use

From Single Agent to Multi-Agent

Complex tasks may require multiple agents.
Example roles:

- planner
- researcher
- writer

Each agent specializes in a task.

Multi-Agent Interaction

Agents communicate via messages.
Examples:

- task delegation
- collaboration
- negotiation

This resembles human teamwork.

Example: Multi-Agent Coding System

Agents collaborate to build software.

- product manager agent
- engineer agent
- testing agent

Workflow:

- specification
- implementation
- testing

Example: Multi-Agent Debate

Agents debate a question.
Process:

1. agent A proposes an answer
2. agent B critiques
3. revisions are generated
4. a judge selects the best answer

Applications

LLM agent systems are used for:

- coding assistants
- research tools
- automation systems
- simulations

Common Failures in LLM Agents

Current agent systems face several issues:

- planning failures
- incorrect tool usage
- hallucinated information
- context limitations

Examples:

- calling the wrong API
- repeating steps
- forgetting previous results

Research Directions in Agent Systems

Important research topics include:

- planning algorithms for LLM agents
- learning tool use
- long-term memory architectures
- multi-agent coordination
- evaluation of agent systems

Future Directions

Future AI systems may involve:

- autonomous AI systems
- improved planning
- cooperative multi-agent systems

Evolution of AI Systems

Machine Learning

$\downarrow$

Deep Learning

$\downarrow$

Transformers

$\downarrow$

Large Language Models

$\downarrow$

AI Agents

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Multi-Agent Systems

Key Takeaways

- Traditional ML models make single predictions.
- LLM agents enable reasoning and multi-step actions.
- Agents combine models, tools, and memory.
- Future AI systems may involve many collaborating agents.